AMA515A Forecasting and applied time series analysis

Chapter 1 : Introduction to forecasting

1. Time series data and forecasts

1.1 The general setup

To build a forecast model for a time series Y_t, a researcher usually collects a sequence of observations from the available data. The data are usually partitioned (see the chart below) in the following fashion:

----------Historical Data---------- →
----sample----------→
$Y_{BEG}(Y_1)$----------→Y_n----------→$Y_{END}(Y_N)$

$\hat{Y}_1$---within-sample forecast--------→ $\hat{Y}_n$ $\hat{Y}_{n+1}$ ----Ex post forecast-----→ $\hat{Y}_N$ $\hat{Y}_{N+1}$-→Ex ante for.

- *Historical data* are provided from time period Y_{BEG} to $Y_{END.}$ We define Y_{BEG} as that time period in which data collection on the variable began and Y_{END} as the most recent observation on the variable. For our purpose Y_{END} might be considered today's observation.
- *The sample period for analysis,* $Y_1,\ldots, Y_n$, is the period over which we will be building or estimating the forecasting model.
- *The ex post period* is defined as the time period from the first observation after the end of the sample period for analysis ($\hat{Y}_{n+1}$) to the most recent observation ($\hat{Y}_N$). The most important characteristic of this time period is that the researcher has available the current values of the time series variable Y_t.
- *The ex ante* forecast period is defined as that time period in which no observations on the time series variable exist.

1.2 Measuring the forecast error

Defn. We define the difference between the actual and the forecast values as the forecast error,

$$e_t = Y_t - \hat{Y}_t$$

The following statistical summary measures of a model's forecast accuracy are defined using the absolute errors:

1. The mean absolute error

$$\text{MAE} = \left(\sum_{1}^{n} |e_t|\right)/n$$

2. The mean of the absolute percentage error

$$\text{MAPE} = \frac{\sum_{1}^{n} \frac{|e_t|}{Y_t}}{n}$$

where e_t is the forecast error in time period t;
Y_t is the actual value in time period t; n is the number of forecast observations in the estimation period.

The following statistical summary measures of a model's forecast accuracy are defined using the squared errors:

1. The mean squared error

$$\text{MSE} = \frac{\sum_{1}^{n} e_t^{\,2}}{n}$$

2. The root mean squared error

$$\text{RMSE} = \sqrt{MSE}$$

3. Theil's U

U is defined as $$U = \frac{\sqrt{\frac{1}{n}\sum_{t=1}^{n}(Y_t - \hat{Y}_t)^2}}{\sqrt{\frac{1}{n}\sum_{t=1}^{n}\hat{Y}_t^{\,2}} + \sqrt{\frac{1}{n}\sum_{t=1}^{n}Y_t^{\,2}}},$$

where $\sum_{t=1}^{n}(Y_t - \hat{Y}_t)^2 =$ the sum of the squared forecast errors, and $\hat{Y}_t$ can also be the within sample forecast, also known as fitted value;

$\sum_{t=1}^{n}\hat{Y}_t^{\,2}$ = the sum of the squared forecast values;

$\sum_{t=1}^{n}Y_t^{\,2}$ = the sum of the squared actual values.

It is clear that U has a value between 0 and 1. A small value of U indicates a good forecasting model.

1.3 Some simple forecast models

(a) The naïve forecast

$$\hat{Y}_{t+1} = Y_t$$

(b) Simple moving average

'k point moving average forecast' = $\hat{Y}_{t+1} = (Y_t + Y_{t-1} + ... + Y_{t-(k-1)})/k$

(c) Linear trend forecast

If we assume a linear regression over time model,

$Y_t = \alpha + \beta t + \varepsilon_{t;}$ then the forecast is given by

$$\hat{Y}_t = a + bt$$

(d) Causal regression model

$$Y_t = \alpha + \beta X_t + \varepsilon_t$$

where Y_t =actual values for the time series variable;

X_t = explanatory variable;

ε_t = random error.

Then $\hat{Y}_t = a + bX_t$

Example: The data below refers to some annual revenues of a certain company.

Year	00	01	02	03	04	05	06
Rev.	375.2	414.2	458.5	511.9	542	588.1	656.4
Year	07	08	09	10	11	12	13
Rev.	722.5	804.2	896.8	957.4	1038.7	1069.3	1167.9

Table 2 gives the actual revenue, naïve, 3-period moving average, & linear trend forecasts for 2006-20133

The linear trend equation for 00 to 05 is $R_t = 374.41 + 42.89 * t$

Note: In the above regression, t runs from 0 to 5.

Table 2 Revenues and forecasts

Year	Actual	Naïve	3-period MA	Trend
06	656.4	588.1	547.3	631.8
07	722.5	656.4	595.5	674.6
08	804.2	722.5	655.7	717.5
09	896.8	804.2	727.7	760.4
10	957.4	896.8	807.8	803.3
11	1038.7	957.4	886.1	846.2
12	1069.3	1038.7	964.3	889.1
13	1167.9	1069.3	1021.8	932.0

1.4 Simple exponential smoothing (for constant process)

Let $X_t = b + \varepsilon_t$

And observations are $x_1, x_2, ..., x_T$. The setup of the simple exponential smoothing (SES) method is :

$$\hat{b}(T) = \hat{b}(T-1) + \alpha[x_T - \hat{b}(T-1)]$$

where $0<\alpha<1$

Rationale : We modify the old estimate(forecast) by some fraction of the forecast error, $x_T - \hat{b}(T-1)$

Let $S_T \equiv \hat{b}(T)$, then we have

$$S_T = S_{T-1} + \alpha[x_T - S_{T-1}],$$

$$\therefore S_T = \alpha x_T + (1-\alpha) S_{T-1}.$$

1.4.1 Properties of simple exponential smoothing forecast

Observe $S_T = S_{T-1} + \alpha[x_T - S_{T-1}]$,

$S_T = \alpha x_T + (1-\alpha)[\alpha x_{T-1} + (1-\alpha)S_{T-2}]$,

$\Rightarrow S_T = \alpha \sum_{k=0}^{T-1} (1-\alpha)^k x_{T-k} + (1-\alpha)^T S_0$

thus S_T is a weighted average of all past observations.

Moreover $(1-\alpha)^T S_0 \to 0$ *as* $T \to \infty$,

Thus $E(S_T) = E[\alpha \sum_{k=0}^{\infty} (1-\alpha)^k x_{T-k}]$

$= b$

S_T is an unbiased estimate of b, thus it is natural to have

$$\hat{x}_{T+\tau} = S_T$$

Rem1 : We frequently use the average of the recent observations as S_0

Rem2 : In terms of updating, exp. Smoothing is a very convenient procedure.

Example: Simple exponential smoothing

The monthly sales of a kind of TV in the past six momths are shown in the following table. Using S_0= 393.13, α=0.1, the smoothed statistics and the one-step ahead forecasts are shown in the following table. Note that S_0 is chosen 'arbitrarily'.

Sequence no.	Actual sales, X_T	Smoothed stat, S_T	Forecast (one-step)
1	330	386.82	393
2	410	389.14	387
3	408	391.02	389
4	514	403.32	391
5	402	403.19	403
6	343	397.17	403

NOTE : S_T is the forecast made at time T, so it is the forecast of $X_{T+1.}$ Many students think S_T is the forecast of X_T, which is **wrong**!

Observe

$$S_1 = 0.1X_1 + 0.9S_0$$

That means $S_1 = 0.1(330) + 0.9(393.13) = 386.82$, and that is the forecast of X_2.

Similarly, $S_2 = 0.1X_2 + 0.9S_1$

So $S_2 = 0.1(410) + 0.9(386.82) = 389.14$, and that is the forecast of X_3.

1.5 Descriptive techniques

1.5.1 Trend and seasonality

Both refer to deterministic variation in the mean of the time series X_t, i.e.,

$$\mu_t = E(X_t)$$

Trend : Long term change in $\mu_t (e.g. \mu_t = \alpha + \beta t)$

Seasonality : Regular periodic variation in $\mu_t (e.g. \mu_t = \mu_{t+4})$

Smoothing may be used to isolate trend and/or seasonality. An alternative is trigonometric regression, e.g., might postulate

$$X_t = \mu + \alpha t + \beta \cos \omega t + \gamma \sin \omega t + Z_t$$

where Z_t is iid $N(0,\sigma^2)$ as possible model for data.

1.5.2 Stochastic trend

Ex. $X_t = X_{t-1} + Z_t$,

$Z_t \sim NID(0, \sigma^2)$

Let $X_0 = 0$, then

$$X_t = \sum_{n=1}^{t} Z_n \sim N(0, t\sigma^2)$$

then var(X_t)= $t\sigma^2$, which shows an increasing trend in the variance, called a stochastic trend.

Define a differencing operation by

$$\nabla X_t = X_t - X_{t-1}$$

then in (2.1),

$\nabla X_t = Z_t \sim N(0,\sigma^2)$, $for all t$.

Thus differencing can eliminate stochastic trend. Similarly, if

$X_t = \mu + \alpha t + Z_t$, $Z_t \sim NID(0,\sigma^2)$. Then

$\nabla X_t = \alpha + (Z_t - Z_{t-1}) \sim NID(\alpha, 2\sigma^2)$, so differencing can eliminate deterministic trend. Similarly , seasonal differencing ,

$\nabla_k X_t = X_t - X_{t-k}$ can be useful.

1.5.3 Causal model and regression over time

(A) Linear model

$Y_t = \alpha + \beta X_t + \varepsilon_t$ ------Causal model

$Y_t = \alpha + \beta t + \varepsilon_t$ -------Trend model

Here ε_t satisfies (i) $\varepsilon_t \sim N(0, \sigma^2)$
(ii) ε_t's are independent

For (ii), testing the presence of autocorrelations gives the famous Durbin-Watson statistic d, where

$$d = \sum_2^n (e_t - e_{t-1})^2 / \sum_1^n e_t^2$$

Now observe

$$\frac{\sum (e_t - e_{t-1})^2}{\sum e_t^2} \cong \frac{2\sum e_t^2 - 2\sum e_t e_{t-1}}{\sum e_t^2}$$

$$\cong 2(1 - \hat{r}_1).$$

Thus 0<d<4, with a value close to zero indicates a +ve correlation, and a value close to 4 indicates a -ve correlation.

Durbin and Watson calculate two more values d_L and d_U to complete the testing procedure. The testing problem is :

$H_0 : \rho = 0$
$H_1 : \rho > 0$ or $\rho < 0$.

Steps: (1) if $d_U < d < 4-d_U$, accept H_0

(2) if $d < d_L$, accept $\rho > 0$
if $d > 4-d_L$, accept $\rho < 0$
(3) if $d_L < d < d_U$ or $4 - d_U < d < 4 - d_L$, then test is inconclusive.

Estimation and testing

This is equivalent to the standard linear regression model case. Thus in

$$Y_t = \alpha + \beta t + \varepsilon_t,$$

$$\hat{\beta} = b = \frac{c_{yt}}{c_{tt}} = \frac{\sum (y_t - \bar{y})(t - \bar{t})}{\sum (t - \bar{t})^2}, \text{ and}$$

$$\hat{\alpha} = a = \bar{y} - b\bar{t}.$$

For testing the slope coefficient β, use $t = \frac{b - \beta}{s_b}$, on n-2 degrees of freedom,

where $s_b = \sqrt{MSE / c_{tt}}$, and MSE= $\sum e_t^2 /(n-2)$.

Forecasting with linear trend

$$y_{t+\tau} = \hat{\alpha} + \hat{\beta}(t + \tau)$$

Prediction interval

$$\hat{y}_t \pm t_{\frac{\alpha}{2}} s_e \sqrt{1 + \frac{1}{n} + \frac{(t_p - \bar{t})^2}{c_{tt}}},$$

where $s_e = \sqrt{MSE}$ and MSE being from linear regression analysis.

(B) Curvilinear model

Tools for detection of nonlinearity:
(1) TS plot
(2) D.W. statistics for residuals from linear regression
(3) Plot of 1st differences --- 1st difference will roughly be a random graph if original data possess only linear trend. If original data come from a quadratic function, then observe

$a+b(t+1)+c(t+1)^2-a - bt- ct^2 = b+2ct+c$, which implies the existence of trend. Thus the presence of trend in the 1st difference indicates a nonlinear model.

Model type:

(I) Polynomial :

$$y_t = a + b_1 t + ... + b_k t^k + \varepsilon_t, t = 1,2,..., n$$

Estimation-- Recall from multiple regression theory, if we let

$\boldsymbol{\beta} = (a\ b_1 \ldots b_k)'$, $\mathbf{X} = \begin{pmatrix} 1 & 1 & ... & 1^k \\ 1 & 2 & ... & 2^k \\ \vdots & & & \\ 1 & t & ... & t^k \end{pmatrix}$, then $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$,

where $\mathbf{y} = (y_1, y_2, \ldots y_n)'$.

(II) Exponential(loglinear) :

$$y_t = ae^{bt}\varepsilon_t,\ \varepsilon_t \sim N(1, \sigma^2)$$
$$\Rightarrow \ln(y_t) = a + bt + \eta_t,\ \eta_t \sim N(0, {\sigma_1}^2)$$
$$\Rightarrow \ln(\hat{y}_t) = \hat{a} + \hat{b}\,t$$

(III) Logistic (S-shaped) :

$$y_t = \frac{L}{1+\exp(\alpha + \beta t + \varepsilon_t)}$$
$$\Rightarrow \hat{y}_t = \frac{L}{1+\exp(\hat{\alpha} + \hat{\beta}\, t)}.$$

Business forecasting

Appendix 1: Prediction interval of trend model

Recall the formula for prediction interval is

$$\hat{y}_t \pm t_{\frac{\alpha}{2}} s_e \sqrt{1+\frac{1}{n}+\frac{(t_p-\bar{t})^2}{c_{tt}}}$$

Using the first 10 obs of Series 1 as example, we have

$Y_t = 2735.9 + 18.67t$

Thus predicted value for 11th obs is 2941.27 approximately.

SS1
2767
2766
2782
2811
2836
2850
2870
2869
2898
2937
2965
2994
3023
3065

Prediction interval is

$$2941.27 \pm t_{8,0.025}\sqrt{109.2}\sqrt{1+\frac{1}{10}+\frac{(11-5.5)^2}{c_{tt}}} =$$

$$2941.27 \pm 2.306\sqrt{109.2}\sqrt{1+\frac{1}{10}+\frac{(11-5.5)^2}{c_{tt}}} = 2941.27 \pm 2.306\sqrt{109.2}\sqrt{1+\frac{1}{10}+\frac{30.25}{82.5}}$$

=(2912.05, 2970.39).